Page 13 Exercises

Let A denote associativity of integers, C denote commutativity of integers. Here, a, b, c, d, x, w, y, z are all integers.

1 LHS $= (a+b)+(c+d)$

$= a+b+c+d$ [A]

$= a+d+b+c$ [C]

$= (a+d)+(b+c)$ [A]

$=$ RHS

2 LHS $= (a+b)+(c+d)$

$= a+b+c+d$ [A]

$= a+c+b+d$ [C]

$= (a+c)+(b+d)$ [A]

$=$ RHS

3 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a+c-b-d$ [C]

$= (a+c)+(-b-d)$ [A]

$=$ RHS

4 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a+c-b-d$ [C]

$= (a+c)+(-b-d)$ [A]

$= (a+c)-(b+d)$ [N5]

$=$ RHS

5 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a-d+c-b$ [C]

$= (a-d)+(c-b)$ [A]

$=$ RHS

6 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= -b-d+a+c$ [C]

$= (-b-d)+(a+c)$ [A]

$= -(b+d)+(a+c)$ [N5]

$=$ RHS

7 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= -b-d+a+c$ [C]

$= -b-d-(-a)-(-c)$ [N4]

$= (-b-d)+(-(-a)-(-c))$ [A]

$= -(b+d)-(-a-c)$ [N5]

$=$ RHS

8 LHS $= ((x+y)+z)+w$

$= x+y+z+w$ [A]

$= x+z+y+w$ [C]

$= (x+z)+(y+w)$ [A]

$=$ RHS

9 LHS $= (x-y)-(z-\omega)$

$= x-y-z-(-\omega)$ [A]

$= x-y-z+\omega$ [N4]

$= x+\omega-y-z$ [C]

$= (x+\omega)-y-z$ [A]

$=$ RHS

10 $(x-y)-(z-\omega)$

$= x-y-z-(-\omega)$ [A]

$= x-y-z+\omega$ [N4]

$= x-z+\omega-y$ [C]

$= (x-z)+(\omega-y)$ [A]

$=$ RHS

11 LHS $= -(\alpha+b+c)$

$= -\alpha-b-c$ [N5]

$= -\alpha+(-b)+(-c)$ [A]

$=$ RHS

12 LHS $= -(\alpha-b-c)$

$= -(\alpha)-(-b)-(-c)$ [N5]

$= -\alpha+b+c$ [N4]

$=$ RHS

13 LHS $= -(\alpha-b)$

$= -(-b+\alpha)$ [C]

~~$= -(-b)-\alpha$ [A]~~ $-(-b)-\alpha$ [A]

$= b-\alpha$ [N4]

$=$ RHS

14 $-2+x=4$

$x = 4+2 = 6$

15 $2-x=5$

$x = 2-5 = -3$

16 $x-3=7$

$x = 10$

17 $-x+4=-1$

$x=5$

18 $4-x=8$

$x=-4$

19 $-5-x=-2$

$x=-3$

20 $-7+x=-10$

$x=-3$

21 $-3+x=4$

$x=7$

22 $a+b=\alpha+c$

$\alpha+b-\alpha=\alpha+c-\alpha$ [subtracting α from both]

$b=c$

$\therefore a+b=\alpha+c \iff b=c$

23 $a+b=\alpha$

$a+b-\alpha=\alpha-\alpha$ [subtracting α from both]

$b=0$

$\therefore \alpha+b=\alpha \iff b=0$ (N1 is proved)

Page 21 Exercises

1 a) $8a^2b^3(27a^4)(2^5ab)$

$= 2^8 3^3 a^7 b^4$

b) $16b^3a^2(6ab^4)(ab)^3$

$= 2^5 3^1 a^6 b^8$

c) $3^2(2ab)^3(16a^2b^5)(24b^2a)$

$= 2^{10} 3^3 a^6 b^{10}$

d) $24a^3(2ab^2)^3(3ab)^2$

$= 2^6 3^3 a^8 b^8$

e) $(3ab)^2(27a^3b)(16ab^5)$

$= 2^4 3^6 a^6 b^8$

f) $32a^4b^5a^3b^2(6ab^3)^4$

$= 2^9 3^4 a^{11} b^{19}$

2 $(a+b)^2(a+b)$

$= (a^2+2ab+b^2)(a+b)$

$= a^3+a^2b+2a^2b+2ab^2+b^2a+b^3$

$= a^3+3a^2b+3ab^2+b^3$

$(a-b)^2(a-b)$

$= (a^2-2ab+b^2)(a-b)$

$= a^3-a^2b-2a^2b+2ab^2+b^2a-b^3$

$= a^3-3a^2b+3ab^2-b^3$

3 $(a+b)^2(a+b)^2$

$= (a^2+2ab+b^2)(a^2+2ab+b^2)$

$= a^4+2a^3b+b^2a^2+2a^3b+4a^2b^2+2ab^3+b^2a^2+2ab^3+b^4$

$= a^4+4a^3b+6a^2b^2+4ab^3+b^4$

$(a-b)^2(a-b)^2$

$= (a^2-2ab+b^2)(a^2-2ab+b^2)$

$= a^4-2a^3b+a^2b^2-2a^3b+4a^2b^2-2ab^3+b^2a^2-2ab^3+b^4$

$= a^4-4a^3b+6a^2b^2-4ab^3+b^4$

4 $(2-4x)^2 = 4 - 16x + 16x^2$

5 $(1-2x)^2 = 1 - 4x + 4x^2$

6 $(2x+5)^2 = 4x^2 + 20x + 25$

7 $(x-1)^2 = x^2 - 2x + 1$

8 $(x+1)(x-1) = x^2 - x + x - 1 = x^2 - 1$

9 $(2x+1)(x+5) = 2x^2 + 10x + x + 5 = 2x^2 + 11x + 5$

10 $(x^2+1)(x^2-1) = x^4 - x^2 + x^2 - 1 = x^4 - 1$

11 $(1+x^3)(1-x^3) = 1 - x^6$

12 $(x^2+1)^2 = x^4 + 2x^2 + 1$

13 $(x^2-1)^2 = x^4 - 2x^2 + 1$

14 $(x^2+2)^2 = x^4 + 4x^2 + 4$

15 $(x^2-2)^2 = x^4 - 4x^2 + 4$

16 $(x^3-4)^2 = x^6 - 8x^3 + 16$

17 $(x^3-4)(x^3+4) = x^6 - 16$

18 $(2x^2+1)(2x^2-1) = 4x^4 - 1$

19 $(-2+3x)(-2-3x) = 4 - 9x^2$

20 $(x+1)(2x+5)(x-2) = (x^2-2x+x-2)(2x+5) = (x^2-x-2)(2x+5)$

~~$= 2x^3 + 5x^2 - 4x^2 - 10x$~~

$= 2x^3 + 5x^2 - 2x^2 - 5x - 4x - 10$

$= 2x^3 + 3x^2 - 9x - 10$

21 $(2x+1)(1-x)(3x+2)$

$= (6x^2 + 4x + 3x + 2)(1-x)$

$= (6x^2 + 7x + 2)(1-x)$

$= 6x^2 - 6x^3 + 7x - 7x^2 + 2 - 2x$

$= -[6x^3 + x^2 - 5x - 2]$

22 $(3x-1)(2x+1)(x+4)$

$= (6x^2 + 3x - 2x - 1)(x+4)$

$= (6x^2 + x - 1)(x+4)$

$= 6x^3 + 24x^2 + x^2 + 4x - x - 4$

$= 6x^3 + 25x^2 + 3x - 4$

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23 $(-1-x)(-2+x)(1-2x)$

$= (2-x+2x-x^2)(1-2x)$

$= (2+x-x^2)(1-2x)$

$= 2-4x+x-2x^2-x^2+2x^3$

$= 2x^3-3x^2-3x+2$

24 $(-4x+1)(2-x)(3+x)$

$= (6+2x-3x-x^2)(-4x+1)$

$= (6-x-x^2)(-4x+1)$

$= -24x+6-4x^2-x+4x^3-x^2$

$= 4x^3-5x^2-25x+6$

25 $(1-x)(1+x)(2-x)$

$= (1-x^2)(2-x)$

$= 2-x-2x^2+x^3$

$= x^3-2x^2-x+2$

26 $(x-1)^2(3-x)$

$= (x^2-2x+1)(3-x)$

$= 3x^2-x^3-6x+2x^2+3-x$

$= -[x^3-5x^2+7x-3]$

27 $(1-x)^2(2-x)$

$= (1-2x+x^2)(2-x)$

$= 2-x-4x+2x^2+2x^2-x^3$

$= -[x^3-4x^2+5x-2]$

28 $(1-2x)^2(3+4x)$

$= (1-4x+4x^2)(3+4x)$

$= 3+4x-12x-16x^2+12x^2+16x^3$

$= 16x^3-4x^2-8x+3$

29 $(2x+1)^2(2-3x)$

$= (4x^2+4x+1)(2-3x)$

$= 8x^2-12x^3+4x-12x^2+2-3x$

$= -[12x^3+4x^2-x-2]$

30 Let population be denoted by P.

According to given conditions

$$P_{(t)} = 50{,}000 \times 2^{t/10}$$

(a) 1970 $\Rightarrow t = 60 \Rightarrow t/10 = 6 \therefore 2^6 = 64$

Hence $P(1970) = (50{,}000)(64) = 3{,}200{,}000$

(b) 1990 $\Rightarrow t = 80 \Rightarrow t/10 = 8 \therefore 2^8 = 512$

Hence $P(1990) = (50{,}000)(512) = 25{,}600{,}000$

(c) 2000 $\Rightarrow t = 90 \Rightarrow t/10 = 9 \therefore 2^9 = 1024$

Hence $P(2000) = (50{,}000)(1024) = 51{,}200{,}000$

31 As we have seen above

(a) $(100{,}000)(4) = 400{,}000$

(b) $(100{,}000)(16) = 1{,}600{,}000$

(c) $(100{,}000)(64) = 6{,}400{,}000$

32 As we have seen above

(a) $(200000)(64) = 12{,}800{,}000$

(b) $(200000)(32) = 6{,}400{,}000$

33 (a) $(25000)(27) = 675000$

(b) $(25000)(81) = 2025000$

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Page 25 Exercises

1 If a is even and b is even, then $a+b$ is even.

From assumptions, $a = 2k$, $b = 2\lambda$.

Then $a+b = 2k + 2\lambda$

$= 2(k+\lambda)$

$a+b = 2\phi$

As $a+b$ can be written as multiple of 2 or as an even integer, $a+b$ must be even.

If a is odd and b is even, then $a+b$ is odd.

From assumptions, $a = 2k+1$, $b = 2\lambda$.

Then $a+b = 2k+1+2\lambda$

$a+b = 2\phi + 1$

$\therefore a+b$ must be odd.

If a is even and b is odd, then $a+b$ is odd. It follows from above.

2 If a is even and b is any positive integer (even or odd), then ab is even.

Let $a = 2k$, $b = 2\lambda + 1$.

$ab = (2k)(2\lambda+1)$

$= 4k\lambda + 2k$

$= 2[2k\lambda + 2k]$

$ab = 2\phi \implies ab$ is even

Let $a = 2k$, $b = 2\lambda$

$ab = (2k)(2\lambda)$

$= 4k\lambda$

$ab = 4\phi \implies ab$ is even

Hence, if a is even and b is any positive integer then ab must be even.

3 Let $\alpha = 2k$.

Then $\alpha^3 = 8k^3 = 2(4k^3) = 2\phi$

Hence α^3 must be even.

4 Let $\alpha = 2k+1$

Then $\alpha^3 = (2k+1)^3 = 8k^3 + 1 + 6k(2k+1)$

$= 8k^3 + 1 + 12k^2 + 6k$

$= 2[4k^3 + 6k^2 + 3k] + 1$

$\alpha^3 = 2\phi + 1$

Hence α^3 must be odd.

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5 Let $n = 2k$. Remember that $(-1)(-1) = 1$.

Then $(-1)^n = (-1)^{2k} = [(-1)(-1)]^k = 1^k = 1$.

Hence, $(-1)^n = 1$ if n is even.

6 Let $n = 2k+1$.

Then $(-1)^n = (-1)^{2k+1} = 1^k(-1) = (1)(-1) = -1$

Hence, $(-1)^n = -1$ if n is odd.

7 Let $m = 2k+1$, $n = 2\lambda+1$.

Then $mn = (2k+1)(2\lambda+1) = 4k\lambda + 2k + 2\lambda + 1 = 2\phi + 1$

Hence mn is odd when m, n are odd.

8 2^4

9 2^3

10 2^5

11 2^2

12 2

13 2^6

14 2^2

15 2^2

16 3

17 3^3

18 3^2

19 3^2

20 3

21 3^0

22 3

23 3^2

24 Let $a - b = 5k$, $x - y = 5\phi$

Then $a + x - b - y = 5k - 5\phi = 5\lambda$

Hence $a + x \equiv b + y \pmod 5$.

Then $ax - by$

$= \alpha x - \alpha y + \alpha y - by$

$= \alpha(x-y) + y(\alpha - b)$

$= \alpha 5\phi + y5k$

$= 5\lambda + 5t$

$= 5\gamma$

Hence $ax \equiv by \pmod 5$.

[Remember a, b, x, y are integers that possess the property $a \equiv b \pmod 5$, $x \equiv y \pmod 5$]

25 The answer follows from question 24. replacing 5 with d.

~~26 Let positive integer be P.~~

(i) $P = 3k$

Then $P^2 = 3\phi$

LHS $= P^2 = (3k)^2 = 9k^2 = 3k$

RHS $= 3\phi$

~~LHS~~. LHS $\equiv$ RHS mod 3.

Hence $P^2 \equiv P \pmod 3$.

(ii) $P = 3k+1$

Then $P^2 = (3k+1)^2 = 9k^2 + 1 + 6k = 3\phi + 1$

Then $P^2 \equiv P \pmod 3$.

(iii) $P = 3k+2$

Then $P^2 = (3k+2)^2 = 9k^2 + 4 + 12k = 3\phi + 4$

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$\in$ → belongs to

26 Let P be a positive integer. Given that P is either of the form $3k$, $3k+1$ and $3k+2$ where k is an integer.

The claim is if P^2 divisible by 3 then P is divisible by 3.

$[\ 3 \mid P^2 \Longrightarrow 3 \mid P\ ]$

(i) $P = 3k$

$P^2 = 9k^2 = 3\phi$

Here $\frac{P^2}{3} \in I$ and $\frac{P}{3} \in I$. So claim holds

(ii) $P = 3k+1$

$P^2 = 9k^2 + 6k + 1 = 3\phi + 1$

Here $\frac{P^2}{3} \notin I$ and $\frac{P}{3} \notin I$. So claim holds

(iii) $P = 3k+2$

$P^2 = 9k^2 + 12k + 4$

Here $\frac{P^2}{3} \notin I$ and $\frac{P}{3} \notin I$. So claim holds.

Hence, $3 \mid P \Longleftrightarrow 3 \mid P^2$ (read as if P square is divisible by 3 if and only if P is divisible by 3 meaning when P square is divisible by 3 so is P.

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Page 39 Exercises

1 (a) $2\alpha = \frac{3}{4}$ (b) $\frac{3\alpha}{5} = -7$ (c) $\frac{-5\alpha}{2} = \frac{3}{8}$

$\alpha = \frac{3}{8}$ $\alpha = \frac{-35}{3}$ $\alpha = -\frac{3}{20}$

2 (a) $3x - 5 = 0$ (b) $-2x + 6 = 1$ (c) $-7x = 2$

$x = \frac{5}{3}$ $x = \frac{5}{2}$ $x = -\frac{2}{7}$

3 (a) $\frac{10}{25} = \frac{2}{5}$ (b) $\frac{3}{9} = \frac{1}{3}$ (c) $\frac{30}{25} = \frac{6}{5}$ (d) $\frac{50}{15} = \frac{10}{3}$ (e) $\frac{45}{9} = \frac{5}{1}$

(f) $\frac{62}{4} = \frac{31}{2}$ (g) $\frac{23}{46} = \frac{1}{2}$ (h) $\frac{16}{40} = \frac{2}{5}$

4 We have $\alpha = \frac{m}{n}$.

From given condition $ab = 1$.

Then $\frac{m}{n} \cdot b = 1 \iff b = \frac{n}{m}$

[So there is a rational number b whose value is exactly $\frac{n}{m}$ such that $\alpha b = b\alpha = 1$. It is the multiplicative inverse]

5 (a) $2x - 7 = 21$

$x = 14$

(b) $3(2x - 5) = 7$

$6x = 22$

$x = \frac{11}{3}$

(c) $(4x - 1)2 = \frac{1}{4}$

$8x - 2 = \frac{1}{4}$

$32x - 8 = 1$

$32x = 9$

$x = \frac{9}{32}$

(d) $-4x + 3 = 5x$

$-9x = 3$

$x = -\frac{1}{3}$

(e) $3x - 2 = -5x + 8$

$8x = 10$

$x = \frac{5}{4}$

(f) $3x + 2 = -3x + 4$

$6x = 2$

$x = \frac{1}{3}$

(g) $\frac{4x}{3} + 1 = 3x$

$4x + 3 = 9x$

$5x = 3$

$x = \frac{3}{5}$

(h) $-\frac{3x}{2} + \frac{4}{3} = 5x$

$\frac{-9x + 8}{6} = 5x$

$-9x + 8 = 30x$

$39x = 8$

$x = \frac{8}{39}$

(i) $\frac{2x - 1}{3} + 4x = 10$

$2x - 1 + 12x = 30$

$14x = 31$

$x = 31/14$

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6 (a) $2x - \frac{3}{7} = \frac{x}{5} + 1$

$5(14x - 3) = (x + 5)7$

$70x - 15 = 7x + 35$

$63x = 50$

$x = \frac{50}{63}$

(b) $\frac{3}{4}x + 5 = -7x$

$3x + 20 = -28x$

$31x = -20$

$x = \frac{-20}{31}$

(c) $\frac{-2}{13}x = 3x - 1$

$-2x = 39x - 13$

$41x = 13$

$x = \frac{13}{41}$

(d) $\frac{4x}{3} + \frac{3}{4} = 2x - 5$

$16x + 9 = 12(2x - 5)$

$16x + 9 = 24x - 60$

$69 = 8x$

$x = \frac{69}{8}$

(e) $\frac{4(1-3x)}{7} = 2x - 1$

$4 - 12x = 14x - 1$

$5 = 26x$

$x = \frac{5}{26}$

(f) $\frac{2-x}{3} = \frac{7}{8}x$

$16 - 8x = 21x$

$29x = 16$

$x = \frac{16}{29}$

7 (a) $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$

$6! = 6 \cdot 120 = 720$

$7! = 7 \cdot 720 = 5040$

$8! = 8 \cdot 5040 = 40320$

$\left[\binom{m}{n} = \frac{m!}{n!(m-n)!} = \frac{m \cdot m-1 \cdot m-2 \cdot \ldots \cdot (m-n)!}{n!\,(m-n)!} = \frac{m \cdot m-1 \ldots m-n+1}{n!} \right]$

(b) $\binom{3}{0} = \frac{3!}{0!\,3!} = 1$ $\binom{3}{3} = \frac{3!}{3!\,0!} = 1$

$\binom{3}{1} = \frac{3!}{1!\,2!} = 3$

$\binom{3}{2} = \frac{3!}{2!\,1!} = 3$

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$\binom{4}{0} = \frac{4!}{0!\,4!} = 1$ $\quad \binom{5}{0} = \frac{5!}{0!\,5!} = 1$ $\quad \binom{5}{5} = \frac{5!}{5!\,0!} = 1$

$\binom{4}{1} = \frac{4!}{1!\,3!} = 4$ $\quad \binom{5}{1} = \frac{5!}{1!\,4!} = 5$

$\binom{4}{2} = \frac{4!}{2!\,2!} = 6$ $\quad \binom{5}{2} = \frac{5!}{2!\,3!} = 10$

$\binom{4}{3} = \frac{4!}{3!\,1!} = 4$ $\quad \binom{5}{3} = \frac{5!}{3!\,2!} = 10$

$\binom{4}{4} = \frac{4!}{4!\,0!} = 1$ $\quad \binom{5}{4} = \frac{5!}{4!\,1!} = 5$

(c) $\binom{m}{n} = \binom{m}{m-n}$

$LHS = \frac{m!}{n!\,(m-n)!} = \frac{m!}{(m-n)!\,(m-(m-n))!} = \binom{m}{m-n} = RHS$

(d) $\binom{m}{n} + \binom{m}{n-1} = \binom{m+1}{n}$

$LHS = \frac{m!}{n!\,(m-n)!} + \frac{m!}{(n-1)!\,(m-n+1)!}$

$= \frac{m!}{n\cdot(n-1)!\cdot(m-n)!} + \frac{m!}{(n-1)!\,(m-n+1)\cdot(m-n)!}$

$= \frac{m!}{(n-1)!\,(m-n)!}\left[\frac{1}{n} + \frac{1}{m-n+1}\right]$

$= \frac{m!}{(n-1)!\,(m-n)!}\left[\frac{m+1}{n(m-n+1)}\right]$

$= \frac{(m+1)!}{n!\,(m+1-n)!}$

$= \binom{m+1}{n} = RHS$

8 Let $a = \frac{m}{n}$ where m, n have no common divisor, $n \neq 0$, $m \neq 0$.

By condition, $\alpha^3 = 2 \iff m^3 = 2n^3 = 2k$. This shows m^3 is even which means m is even.

Also, $2n^3 = 2k$ ➡

8 Let be a a positive rational number, we can write $\alpha = \frac{m}{n}$. Also $a^3 = 2$.

Hence $m^3 = 2n^3$. So $m = 2k$.

So $(2k)^3 = 2n^3$. So $n = 2\phi$.

Hence m, n must have 2 as a common divisor. But m and n have no common divisor. Hence our assumption that a is a rational number must be false.

9 Similarly we can show that there is no positive rational number a such that $\alpha^4 = 2$.

10 Let a be a positive rational number so $a = \frac{m}{n}$.

So $m^2 = 3n^2$. - From problem 26 on page 26 [1, §4], that square of a integer divisible by 3 then so is the integer. Hence, $m = 3k$.

Thus $n = 3\phi$.

By similar argument a cannot be rational.

If $\sqrt{2}$ is rational then its square must be rational too.

$a = \sqrt{2} \iff \alpha^2 = 2$

We know from previous theorem 4 that this is not the case.

11 (a) 1 gives the square 1 and 2 gives the square 4

So the number whose square gives 2 must be between 1 and 2

1.4 gives the square 1.96 and 1.5 gives the square 2.25

So the number whose square gives 2 must between 1.4 and 1.5

Repeating gives us 1.414.

(b) We find the number to be 1.4142.

12 (a), (b) We get 1.732.

13 (a), (b) we get 2.236.

14 (a), (b) 1.259.

15 (a), (b) 1.442.

16 We start with 6 grams. It halves every 3 minutes. Using the previous logic used in exercises of [1, § 3],

$W(t) = 6\left(\frac{1}{2}\right)^{t/3}$ where t is in minutes and $W(t)$ in grams.

(a) $W(3) = \frac{6}{2} = 3g$

(b) $W(27) = 6\left(\frac{1}{2}\right)^{9} \sim 0.0117\,g$

(c) $W(36) = 6\left(\frac{1}{2}\right)^{12} \sim 0.00146\,g$

[You may use fractions for exact answer]

17, 18, 19, 20, 21 can be solved in a similar way.